# The bus route

## Optimization · Level 3 · Lesson 3



# In this lesson you will

- Explain why the number of routes explodes.
- Write nearest neighbour in Python and improve a route with 2-opt.
- Judge a route by its length and its longest ride.

# Key ideas

- n! ÷ 2 routes
- Nearest neighbour: fast, greedy
- 2-opt: reverse a stretch
- A local best
- Longest ride, both directions

# Activity: A fairer round trip

1. Map (5 min): Five places and their distances.
2. Build (5 min): Nearest neighbour; add it up.
3. Improve (5 min): Find a crossing; swap it.
4. Ride (5 min): Longest ride, both directions.

**Distances can be minutes of walking. Use made-up names if you prefer not to show real places.**

# Let’s discuss

1. Why does nearest neighbour often end with a long leg?
2. How can 2-opt stop before the best route?
3. Should the bus ever skip a stop to save fuel?

# Exit ticket

- Draw a route with a crossing + the route after one 2-opt swap.